

Practical Quantum Subspace Expansion: Stability, Noise, and Scaling

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Abstract

1 Introduction

- 1.1 The Quantum Computing Edge in Quantum Chemistry
- 1.2 Quantum Subspace Expansion for Excited States
- 1.3 Numerical Instability in Practical QSE
- 1.4 Generalized Eigenvalue Conditioning
- 1.5 Current QSE Studies and Contributions of This Work

2 Results and Discussion

- 2.1 Effective Dimensions of the QSE Subspace
- 2.2 Subspace Completeness Beyond Ansatz Choice
- 2.3 Finite-Shot Effects on Generalized Eigenvalue Stability
- 2.4 Scaling of QSE and Future Adaptive Strategies

3 Computational Methods

- 3.1 Molecular Systems and Active Spaces
- 3.2 VQE Reference States and Ansatz Families
- 3.3 QSE Operator Pools and Projected Matrices
- 3.4 Generalized Eigenvalue Solvers and Stability Filters
- 3.5 Finite-Shot Matrix Estimation
- 3.6 Accuracy, Matching, and Scaling Metrics

4 Conclusions

Acknowledgements

Data Availability